DIGIBRIDGE

Digital Divide App

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LSIS 5171

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Executive Summary

In our modern day, much of life requires access to technology, internet access, and digital and media literacy. As of 2025, 23.9 million people in the U.S. remained offline which is around 6.9 percent of the US. (Kemp, 2025). There are multifactorial reasons why people remain offline which is clustered into one umbrella term—Digital Divide. Divide encompasses lack of access to technology and services such as Wi-Fi and computers, how much the services and technology costs, the quality and suitability of the devices, and the digital literacy to navigate the online world. (GovFacts, 2025). The DigiBridge project was designed as an offline-first mobile and web system that provides three essential capabilities: trusted access-point discovery, adaptive digital literacy tools, and a peer-to-peer community mode that functions without internet access. These features directly support national digital equity goals and the needs of community centers, libraries, and individuals with limited connectivity.

During the expansion of the project, there were significant additions added to help users further and become more inclusive. When creating the needs assessment, it verified that verified Wi-Fi access points, offline learning tools, multilingual and accessible interfaces, and low-resource operation were necessary to provide the most inclusive services and support online and offline spaces.

The feasibility assessments demonstrated that DigiBridge is achievable using established technologies such as offline-first architecture, Bluetooth LE, and Wi-Fi Direct. Economic feasibility is strengthened by long-term reductions in server dependency, while operational feasibility is supported by the system's low-resource design and minimal training requirements. Political and legal feasibility are reinforced by strong public support for digital inclusion initiatives and the privacy benefits of offline-first design.

Key accomplishments in this phase include the development of the use case, communication plan, Gantt chart, system scope, and research. The use case outlines how users will access offline maps, view trusted Wi-Fi locations, and complete literacy modules, while the communication plan establishes a structured process for weekly updates, bi-weekly client meetings, prototype reviews, and emergency communication. These deliverables ensure transparency, alignment, and predictable progress throughout the project lifecycle.

Several important conclusions emerged from this phase. First, offline-first design is not only feasible but essential for the target population. Second, peer-to-peer communication significantly enhances community resilience by enabling information sharing without internet access. Third, multilingual and accessibility features must be integrated at the same time as other features as they are imperative to the system's mission.

The project also identified several challenges and anticipated risks. The complexity of implementing offline-first sync logic and peer-to-peer communication may extend development time. Ensuring accurate, up-to-date access-point data requires ongoing maintenance and community partnerships. Additionally, designing adaptive literacy modules that function entirely offline presents instructional and technical challenges. These risks will be mitigated through phased implementation, continuous testing, and collaboration with community organizations.

It is recommended to adopt a phased implementation strategy to reduce complexity and deliver early value, beginning with the Access-Point Locator before expanding to literacy modules and Community Mode.

Overall, the analysis confirms that the proposed solution is feasible, impactful, and aligned with long-term digital equity goals. The next phase will focus on detailed system design, UI development, and the implementation of the first major feature: the Access-Point Locator.

Client Needs, Proposed Alternatives, Proposed Project Vision

Overview of Client and Client Needs

Client description**:** The client is the population affected by digital inequity, community organizations, libraries, and local service providers who support the libraries, community organizations.

Client Needs Assessment

- **Methods used to assess client needs:**
    - Research and observation: NDIA, PEW Research Center, and DataReportal helped give statistics.
    - Environment scanning: Searching existing tools and existing initiatives and viewing the gaps in offline-first tools.
    - Preliminary stakeholder identification: Considering the needs of individuals in the digital divide, community centers, and public service
- **Identified needs were:**
    - **Reliable access point information**: Users need verified, safe locations where free Wi-Fi is available.
    - **Offline digital literacy tools:** Many users will not have constant connectivity to rely on, so literacy modules must work offline.
    - **Multilingual and accessible design:** Users vary widely in language, ability, and familiarity with technology.

- Low-resource operation: Devices may be older, low-power, or borrowed.
- **Security and privacy:** Users may be in vulnerable populations who need safe, non-exploitative digital environments.
- **Community-based communication:** Users may need to share information locally without internet access.

These findings confirm the need for a system that supports offline learning, provides access point information, and allows safe, low-resource systems for participation.

Proposed Alternatives: Scope and Complexity

1. **Alt 1: Basic Access-Point Locator App**
   - **Scope**: Limited to mapping free Wi-Fi locations
   - **Complexity**: Low.
   - **Advantages**: Quick to build, maintainable, solves an issue.
   - **Disadvantages**: Doesn't address digital literacy, offline use, or community communication.
2. **Alt 2: Offline-First Digital Literacy App**
   - **Scope:** Focused on literacy modules that work without internet.
   - **Complexity:** Moderate.
   - **Advantages:** Addresses a core barrier to digital inclusion.
   - **Limitations:** Does not solve access-point discovery or community communication.
3. **Alt 3: DigiBridge Integrated System** (Proposed Solution):

- **Scope:** Comprehensive system including offline-first literacy tools, access-point mapping, multilingual support, accessibility features, and a peer-to-peer community mode.
- **Complexity:** High, involving offline-first architecture, Bluetooth LE, peer-to-peer Wi-Fi, adaptive tutorials, and cross-platform design.
- **Advantages:** Addresses all identified needs, scalable, supports both online and offline contexts, and empowers users to participate digitally even with limited resources.
- **Limitations:** May take a while.

The integrated DigiBridge system was selected because it provides the most complete solution to the digital divide problem and aligns with long-term digital equity goals.

Final Solution

The final solution is DigiBridge, an offline-first mobile and web application that provides digital literacy tools, verified access-point locations, and a peer-to-peer community mode for communication without internet access. The system will use Bluetooth LE and Wi-Fi Direct to allow nearby users to share maps, resources, and information.

- **Problems addressed:**
    - Lack of reliable internet access
    - Limited digital literacy
    - Difficulty locating safe Wi-Fi
    - Barriers to participating in online services
    - Need for low-resource, secure, multilingual tools
- **Stakeholders and Users:**

    - **Primary Stakeholders**: Individuals affected by the digital divide, community centers, libraries, digital inclusion organizations.
    - **End Users**: People with limited internet access, low digital literacy, or older devices.
    - **Secondary Stakeholders**: Local governments, nonprofits, educators, and public service providers.
- **System Benefits:**
    - **For Users**: Offline learning, safe access-point discovery, community support, and improved digital confidence.
    - **For Organizations:** Better outreach, improved service delivery, and data-informed digital inclusion strategies.
    - **For Communities:** Stronger local networks, improved access to essential services, and reduced digital inequity.

Feasibility Assessment

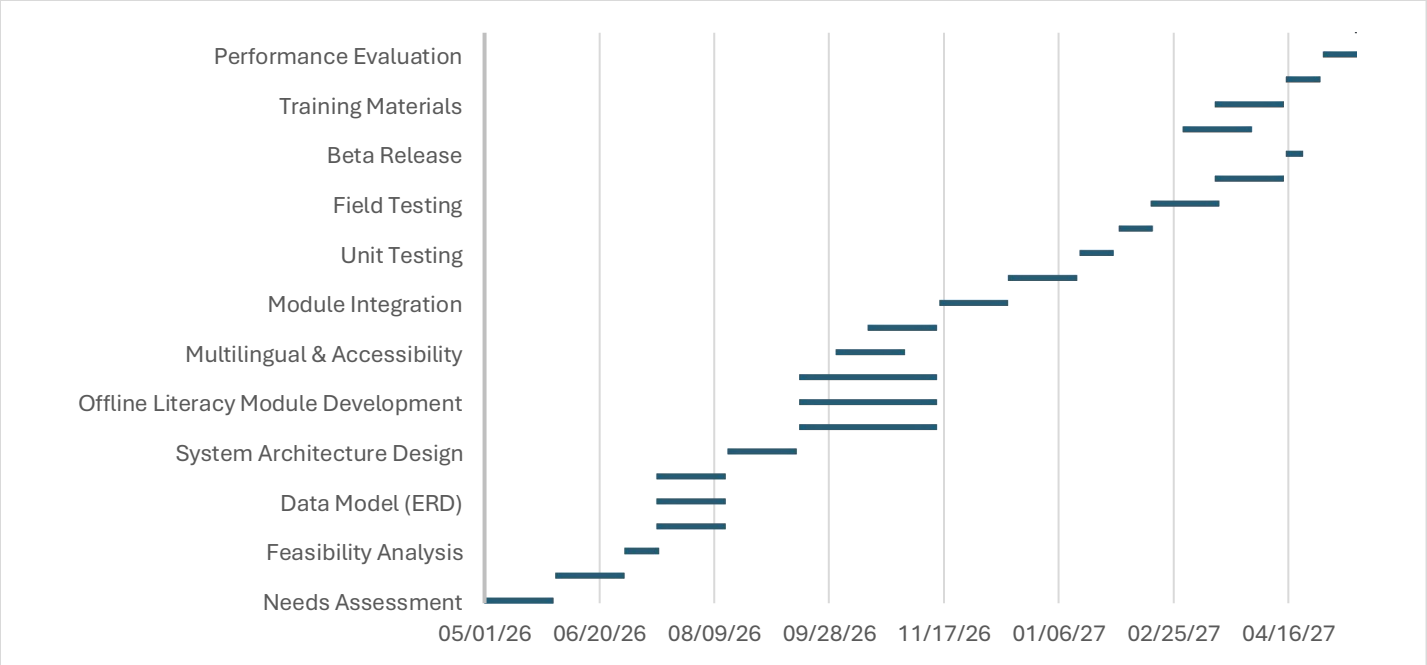
- **Technical Feasibility:** The system relies on established technologies such as offline-first architecture, Bluetooth LE, Wi-Fi Direct, and cross-platform frameworks. These technologies are well-documented and feasible to implement
- **Economic Feasibility:** Initial development costs may be significant due to the complexity of offline-first design and peer-to-peer communication. However, long-term benefits include reduced dependency on constant server infrastructure and increased community impact.

- **Operational Feasibility:** The system is designed for low-resource environments and minimal training. Community organizations can easily adopt it, and users can learn at their own pace.
- **Scheduling Feasibility:** A phased rollout is recommended:
    - Access-point locator
    - Offline literacy modules
    - Community mode
    - This staged approach keeps the project manageable.
- **Legal and Contractual Feasibility:** The system must comply with privacy laws, especially regarding location data and user communication. Offline-first design reduces exposure to data breaches.
- **Political Feasibility:** Digital inclusion initiatives have strong support from public agencies and nonprofits. The project aligns with national digital equity goals.

Activities and Gantt Chart

- Needs Assessment
- Requirements Gathering
- Feasibility Analysis
- Use Case Development
- Data Model (ERD)
- UI/UX Wireframes
- System Architecture Design
- Access-Point Locator Development

- Offline Literacy Module Development
- Peer-to-Peer Communication
- Multilingual & Accessibility
- Security & Privacy Implementation
- Module Integration
- Sync Logic (Offline/Online)
- Unit Testing
- Usability Testing
- Field Testing
- Bug Fixing & Refinement
- Beta Release
- Documentation
- Training Materials
- Feedback Collection
- Performance Evaluation
- Final Revisions



Use Case

Use Case: Accessing Offline Digital Literacy Tools and Finding Trusted Wi-Fi Locations

Primary Actor

Community member with limited internet access.

Stakeholders & Interests

- **User:** Wants safe Wi-Fi locations and offline learning tools.
- **Community Centers/Libraries:** Want to support digital inclusion.
- **DigiBridge System:** Must provide accurate, secure, low-resource functionality.

Preconditions

- User has installed the DigiBridge app or accessed the website at least once.
- User has granted location permission (optional but recommended).
- Offline map and literacy modules have been downloaded.

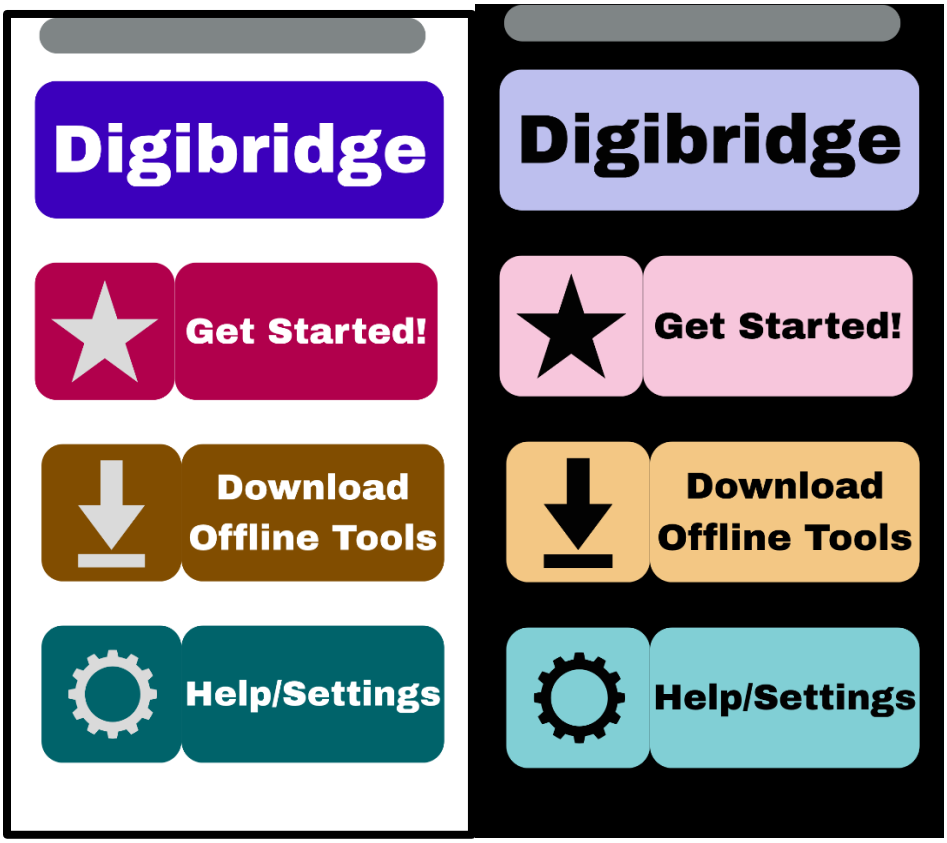
Postconditions

- User successfully views nearby trusted Wi-Fi access points.
- User completes or begins a digital literacy module offline.
- System syncs progress when internet becomes available.

Use Case Scenario

1. User opens the DigiBridge app.
2. System loads offline map and cached access point data.
3. User selects **“Find Wi-Fi Near Me.”**
4. System displays verified access points with distance and safety indicators.
5. User selects **“Digital Literacy Tools.”**
6. System presents adaptive modules based on the user’s skill level.
7. User completes a lesson offline.
8. When the device later connects to Wi-Fi, the system syncs progress and updates access point accuracy.

Forms/Reports/Dialogues



I made a light mode and dark mode of the starter page UI on Canva for an iPhone. I know it needs to be on most technology, but I can't do all of that in two weeks without help. It shows colors chosen for an accessibility-friendly color palette and also, buttons with symbols because the color palette isn't friendly for people with Achromatopsia. I consulted WCAG and some online checkers to plan out the UI and used settings on my laptop to make sure it is easy to read regardless of color blindness levels.



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The plan for Find *Wi-Fi Near Me* was similar in intention. I used my nearby school (I live in Fayetteville, North Carolina), Fayetteville State University, to list how the function of finding Wi-Fi would be like. I actually had to code and screenshot this map picture. It is from an open-source Geographic Information System—GIS—called Leaflet, created by Volodymyr Agafonkin, predominately made in JavaScript and can be implemented into most programming languages. I used R programming for convenience (because I know the language well enough), but I could've used WinForms, Python, or any other programming language. The user has options to refresh offline data when they go to a source of Wi-Fi, they can download a map—Leaflet has the ability

to be downloaded offline—and report an issue, which requires Wi-Fi. A pop-up will come up when they need help to not clutter the iPhone screen.

Communication Plan

To ensure transparency and alignment with the client, the DigiBridge development team will use the following communication structure:

1. Weekly Status Updates

- Delivered via email every Friday.
- Includes: completed tasks, upcoming tasks, risks, blockers, and decisions needed.
- Attachments: updated Gantt chart, sprint notes, and any new mockups.

2. Bi-Weekly Client Meetings (Virtual)

- 30–45 minutes via Zoom or Teams.
- Agenda:
    - Review progress
    - Demonstrate new features
    - Discuss feedback
    - Adjust priorities

3. Issue Tracking & Documentation

- All tasks, bugs, and enhancements tracked in a shared project board (Trello, Jira, or Teams Planner).
- Client have read-only access to monitor progress in real time.

4. Prototype & UI Review Sessions

- Conducted at the end of each major phase (Design, Development, Testing).
- Client reviews wireframes, UI screens, and workflow diagrams.

- Feedback incorporated into the next sprint.

5. Emergency Communication

- Critical issues communicated within 24 hours via email and phone.
- Examples: security concerns, major delays, unexpected technical blockers.

6. Final Deliverables Review

- Client receives final documentation, user guides, and training materials.
- A final meeting is held to walk through the system and confirm acceptance.

Proposed Implementation/Maintenance Plan

The system will be rolled out in phases to make sure that the development is manageable for developers and users as well as the development in general.

Implementation Plan

Phase 1 – Access-Point Locator

This phase makes up the heart of the software and establishes offline-first infrastructure for the users to connect safely online.

Objectives:

- Built the architecture (offline-first)
- Implement location permissions and an easy-to-read privacy statement
- Load offline map tiles and cached access-point data (via Leaflet)
- Display verified Wi-Fi locations with safety indicators. (✓,✕)
- Conduct initial usability testing with community partners

Key Activities:

- Develop map cache and a way to store maps
- Create access-point database schema
- Implement UI for “Find Wi-Fi Near Me”
- Integrate safety indicators and metadata
- Perform early field testing

Deliverables:

- Offline map engine
- Access-point locator module
- Basic UI screens
- Initial test report

Related expansion: Access-Point UI

Phase 2 – Offline Digital Literacy Modules

Focuses on the accessible and multilingual aspect of the digital literacy tools that work without internet access.

Objectives:

- Build adaptive learning engine
- Add multilingual content packs
- Implement accessibility features (contrast, text scaling, screen reader support)
- Enable offline lesson progress tracking
- Implement sync logic for when internet becomes available

Key Activities:

- Develop literacy module templates
- Create adaptive difficulty algorithm

- Build offline content
- Implement progress-sync service

Deliverables:

- Digital literacy module library
- Adaptive learning system
- Offline content packs
- Sync engine

Related expansion: Digital Literacy UI

Phase 3 — Peer-to-Peer Community Mode (Advanced Release)

This is the final phase; it introduces offline communication features using Bluetooth LE and Wi-Fi Direct.

Objectives:

- Implement Bluetooth LE device discovery
- Implement Wi-Fi Direct / Multipeer Connectivity
- Enable sharing of maps, resources, and announcements
- Build Community Mode UI
- Conduct field testing with real users

Key Activities:

- Develop P2P communication protocol
- Build resource-sharing interface
- Implement local announcements and safety alerts
- Conduct multi-device testing
- Integrate user feedback

Deliverables:

- Community Mode engine
- P2P communication module
- Resource-sharing UI
- Field test report

Related expansion: Community Mode UI

Maintenance Plan

Routine Maintenance (Monthly)

Ensures the system remains accurate, stable, and usable in low-resource environments.

- Update access-point database
- Refresh literacy content
- Patch UI bugs
- Review sync logs for errors
- Validate offline map accuracy

Related expansion: Monthly Maintenance Checklist

Security & Privacy Maintenance (Quarterly)

Aligned with NIST and OWASP guidance referenced in the project documents.

- Apply OWASP Mobile Security updates
- Review Bluetooth/Wi-Fi Direct vulnerabilities
- Audit local data storage
- Update encryption libraries
- Validate compliance with privacy laws

Related expansion: Security Plan

Performance & Device Compatibility (Bi-Annual)

Ensures DigiBridge remains functional on older, low-resource devices.

- Test performance on low-end smartphones and borrowed laptops
- Optimize storage footprint
- Update cross-platform frameworks
- Improve caching and offline-sync efficiency

Community Feedback Loop (Continuous)

Matches the Communication Plan's requirement that feedback be incorporated into each sprint.

- Collect user feedback through offline forms
- Review reports from libraries and community centers
- Prioritize improvements in sprint planning
- Update UI based on accessibility findings

Related expansion: Feedback Form UI

Documentation & Training (Annual)

Supports long-term adoption by organizations and nonprofits.

- Update user guides and training materials
- Refresh onboarding tutorials
- Provide updated training for community partners
- Review documentation for clarity and accessibility

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